# PHASETRACKER

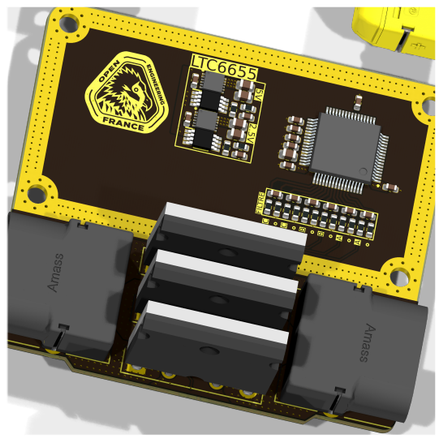
# 2026-10-10

## Variant:

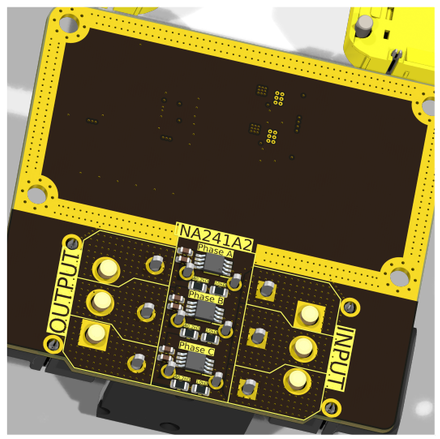
## Rev + (Unreleased)


| Page | Index | Page | Index | Page | Index | Page | Index |
|---|---|---|---|---|---|---|---|
| 1 | Cover Page | 11 | Error: XML File not found | 21 | Error: XML File not found | 31 | Error: XML File not found |
| 2 | Error: XML File not found | 12 | Error: XML File not found | 22 | Error: XML File not found | 32 | Error: XML File not found |
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| 10 | Error: XML File not found | 20 | Error: XML File not found | 30 | Error: XML File not found | 40 | Error: XML File not found |


## TOP VIEW




## BOTTOM VIEW




## NOTES

3-phase current and voltage sensing: INA241A2 + AD7606C-18

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 10-Oct-2026

## DESIGN CONSIDERATIONS

DESIGN NOTE:
Current: 2.5 V - 20 mV/A x I(J1->J2).
Voltage: 0.198 V/V.

DESIGN NOTE:
No Power-Good: rails within 0.1 % after 71 ms.

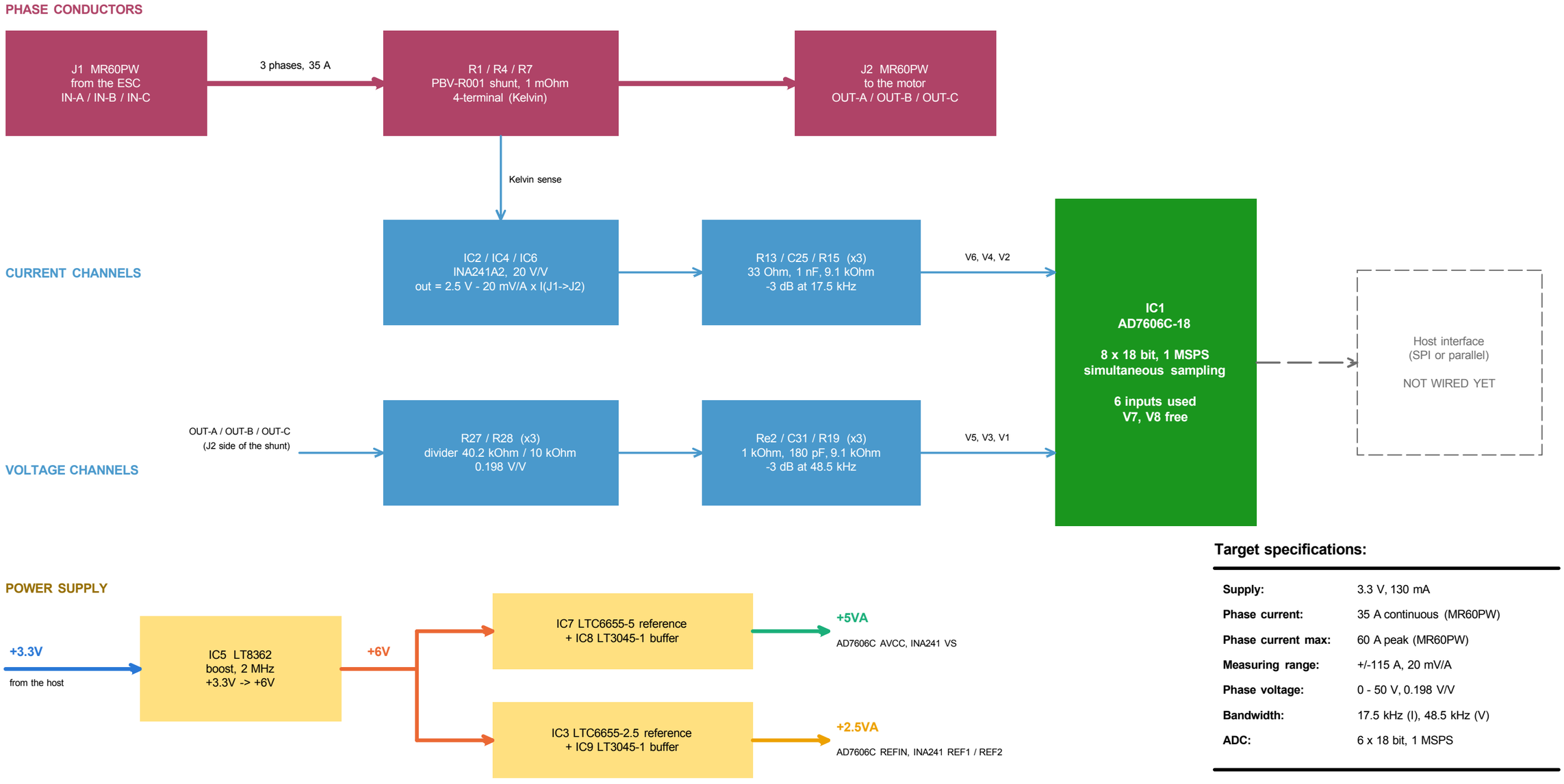
DESIGN NOTE:
AD7606C digital pins and +3.3V input not wired yet.

DESIGN NOTE:
+2.5VA: R12 = 49.9k gives 2.55 V, 24.9k gives 2.50 V. See README.

LAYOUT NOTE:
Kelvin-route the shunt sense pads to each INA241.

| | | | | |
|---|---|---|---|---|
| | Comments: | Company: OPEN THRUST LAB | Variant: | Git Hash: 7703dc5 |
| | | Board Name: PHASETRACKER | Project Name: OPEN_PHASE | |
| | Sheet Title: OPEN_PHASE - PHASETRACKER | File Name: OPEN_PHASE.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-06 / Revision: + (Unreleased) |
| | Sheet Path: / | | Reviewer: | Size: A3 / Sheet: 1 of 10 |

# [2] Block Diagram



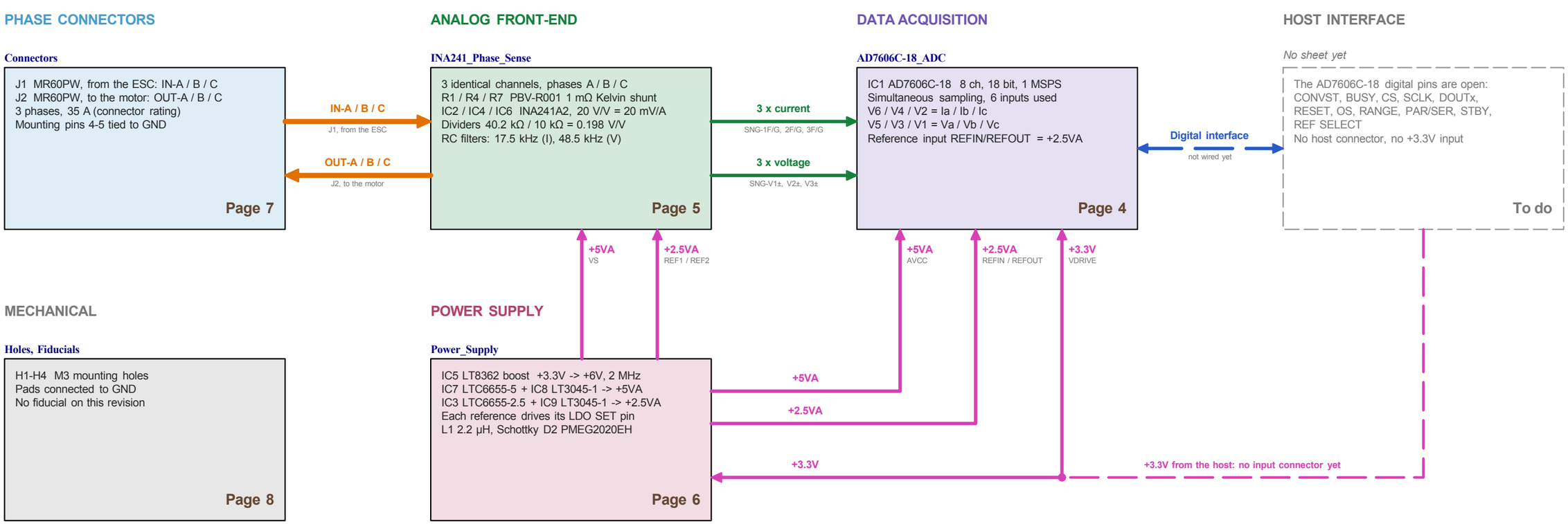

## Target specifications:

| | |
|---|---|
| **Supply:** | 3.3 V, 130 mA |
| **Phase current:** | 35 A continuous (MR60PW) |
| **Phase current max:** | 60 A peak (MR60PW) |
| **Measuring range:** | +/-115 A, 20 mV/A |
| **Phase voltage:** | 0 - 50 V, 0.198 V/V |
| **Bandwidth:** | 17.5 kHz (I), 48.5 kHz (V) |
| **ADC:** | 6 x 18 bit, 1 MSPS |

| | | | | |
|---|---|---|---|---|
| Comments: | Company: OPEN THRUST LAB | | Variant: | Git Hash: 7703dc5 |
| | Board Name: PHASETRACKER | | Project Name: OPEN_PHASE | |
| Sheet Title: Block Diagram | File Name: Block Diagram.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-06 | Revision: + (Unreleased) |
| Sheet Path: /Block Diagram/ | | Reviewer: | Size: A3 | Sheet: 2 of 10 |

# [3] Project Architecture

*Sheet hierarchy and signal flow. Click a block or its page number to open the sheet.*




## LEGEND

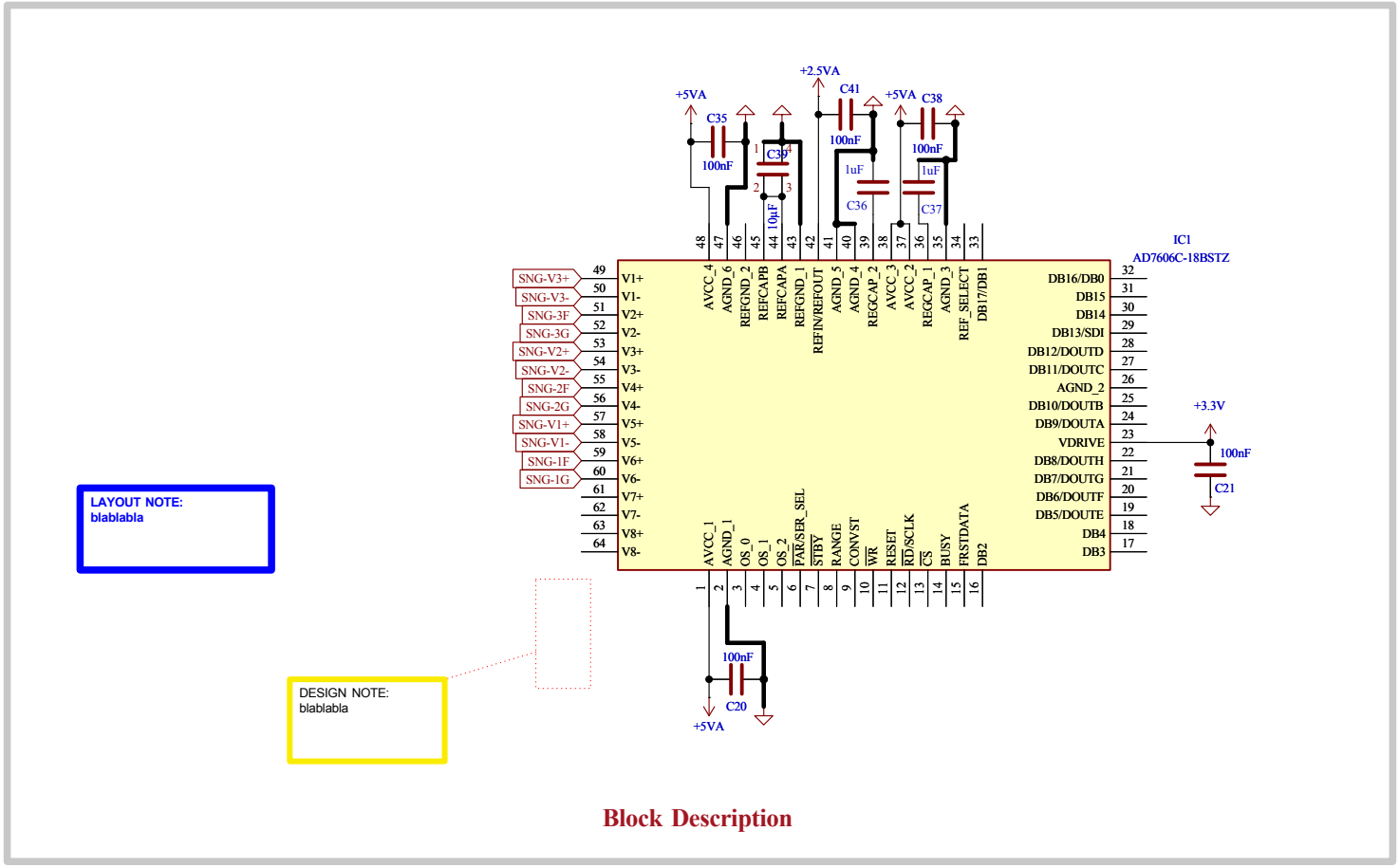
- Phase conductor, 35 A
- Power rail
- Analog signal, differential pairs
- Digital interface, not wired yet
- Supply input, no connector yet

## DOCUMENT INDEX


[1] Title Page
[2] Block Diagram
[3] Project Architecture
[4] AD7606C-18 ADC
[5] INA241 Phase Sensing
[6] Power Supply - Analog Rails
[7] Connectors
[8] Holes, Fiducials
[9] Power Sequencing
[10] Revision History


| | | | | |
|---|---|---|---|---|
| Comments: | Company: OPEN THRUST LAB | | Variant: | Git Hash: 7703dc5 |
| | Board Name: PHASETRACKER | | Project Name: OPEN_PHASE | |
| Sheet Title: Project Architecture | File Name: Project Architecture.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-06 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/ | | Reviewer: | Size: A3 | Sheet: 3 of 10 |

# [4] AD7606C-18
# - ADC




**Block Description**

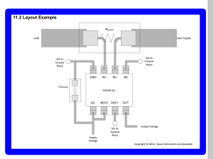
| | | | | |
|---|---|---|---|---|
| Comments: AN5346; STM32G474 Datasheet p.81; J. Pieper ADC investigation | Company: OPEN THRUST LAB | | Variant: | Git Hash: 7703dc5 |
| | Board Name: PHASETRACKER | | Project Name: OPEN_PHASE | |
| Sheet Title: AD7606C-18 - ADC | File Name: AD7606C-18 - ADC.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-05 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/AD7606C-18_ADC/ | | Reviewer: | Size: A4 | Sheet: 4 of 10 |

# [5] INA241 - Phase Sensing

**The INA240 device is a voltage-output, current-sense amplifier with enhanced PWM rejection that can sense drops across shunt resistors over a wide common-mode voltage range from -4 V to 80 V, independent of the supply voltage. The negative common-mode voltage allows the device to operate below ground, accommodating the flyback period of typical solenoid applications. Enhanced PWM rejection provides high levels of suppression for large common-mode transients (ΔV/Δt) in systems that use pulse width modulation (PWM) signals (such as motor drives and solenoid control systems). This feature allows for accurate current measurements without large transients and associated recovery ripple on the output voltage.**
**This device operates from a single 2.7-V to 5.5-V power supply, drawing a maximum of 2.4 mA of supply current. Four fixed gains are available: 20 V/V, 50 V/V, 100 V/V, and 200 V/V. The low offset of the zero-drift architecture enables current sensing with maximum drops across the shunt as low as 10-mV full-scale. All versions are specified over the extended operating temperature range (-40°C to +125°C), and are offered in an 8-pin TSSOP and 8-pin SOIC packages.**

DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V (fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

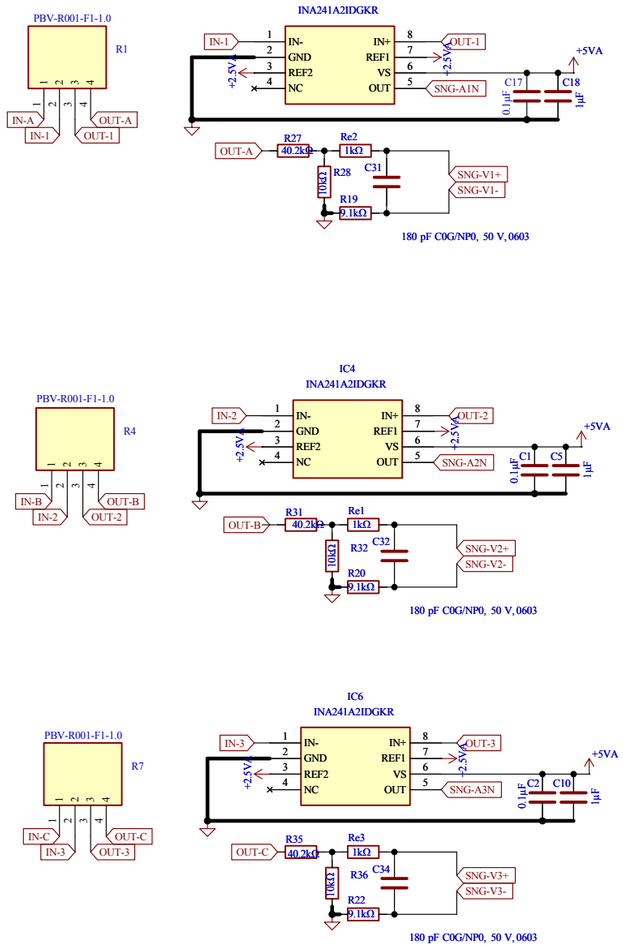
DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V



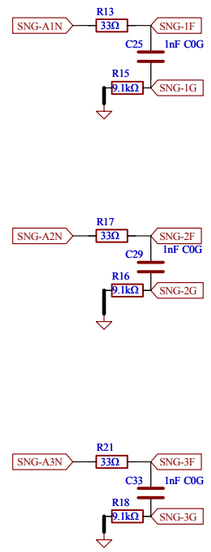

## Description

**Vout≈Vref+(Gain)×(I×Rshunt)**

## Shouse your INA240A...




## INA241A + Shunt




## FILTRE

## Test Point

| | | | | |
|---|---|---|---|---|
| Comments: **References:** Flexible I/O worked examples; Flexible I/O source configuration | Company: OPEN THRUST LAB | | Variant: | Git Hash: 7703dc5 |
| | Board Name: PHASETRACKER | | Project Name: OPEN_PHASE | |
| Sheet Title: INA241 - Phase Sensing | File Name: INA241 - Phase Sensing.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-05 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/INA241_Phase_Sense/ | | Reviewer: | Size: A3 | Sheet: 5 of 10 |

# [6] Power Supply
# - Analog Rails

DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V
(fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

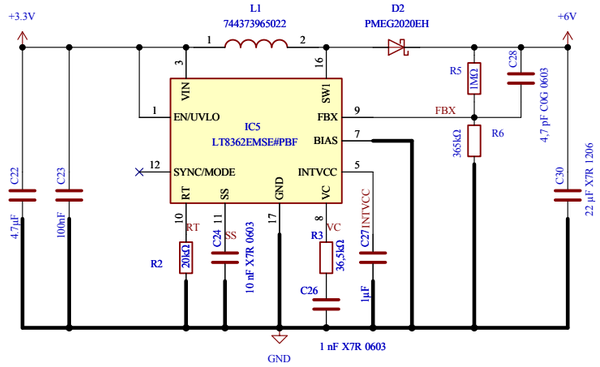
DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V

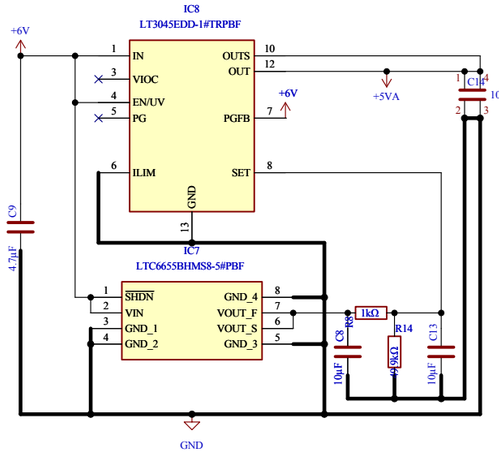
**LAYOUT NOTE:**
blabla

## Description

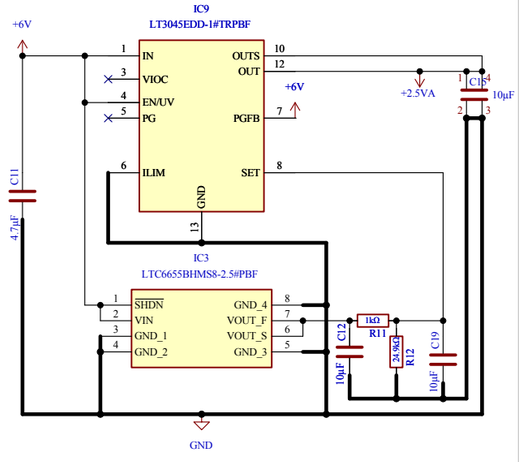
**Vout≈Vref+(Gain)×(I×Rshunt)**

## Shouse your INA240A...




## 5VA précision Alimentation




## 2.5VA précision Alimentation

Don't panic

it's just a regular capacitor with a slightly modified footprint to better cover the 1 MHz and higher bands.

Non-critical detail.

## Test Point

| | | | | |
|---|---|---|---|---|
| Comments:<br>**References:**<br>Flexible I/O worked examples<br>Flexible I/O source configuration | Company:<br>**OPEN THRUST LAB** | | Variant: | Git Hash:<br>7703dc5 |
| | Board Name:<br>**PHASETRACKER** | | Project Name:<br>**OPEN_PHASE** | |
| Sheet Title:<br>Power Supply - Analog Rails | File Name:<br>Power Supply - Analog Rails.kicad_sch | Designer:<br>SIEBERT DIMITRY | Date:<br>2026-10-06 | Revision:<br>+ (Unreleased) |
| Sheet Path:<br>/Project Architecture/Power_Supply/ | | Reviewer: | Size:<br>**A3** | Sheet:<br>**6** of **10** |

# [7] Connectors

**MR60PW stat :**

- **Current: 35 A max (with specified temperature rise < 85°C)**
- **Voltage: 600 V DC**
- **Insulation resistance: ≥ 2000 MΩ**
- **Contact resistance: ≤ 1.0 mΩ**
- **Mating cycles (lifetime): 100 cycles**
- **Salt spray: 48 h**
- **Ingress protection rating: IP40**
- **Operating temperature: -20°C to 120°C**
- **Flammability rating: UL94 V-0 (0.8 mm wall thickness)**

**MR60PW**

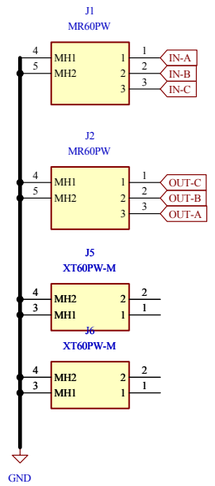
DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V
(fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V

**LAYOUT NOTE:**
**blabla**

**Vout≈Vref+(Gain)×(I×Rshunt)**

**Shouse your INA240A...**




**CONNECTEUR**

**FILTRE**

**Test Point**

| | | | | |
|---|---|---|---|---|
| Comments: **References:** Flexible I/O worked examples; Flexible I/O source configuration | Company: **OPEN THRUST LAB** | | Variant: | Git Hash: 7703dc5 |
| | Board Name: **PHASETRACKER** | | Project Name: **OPEN_PHASE** | |
| Sheet Title: Connectors | File Name: Connectors.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-05 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Connectors/ | | Reviewer: | Size: **A3** | Sheet: **7** of **10** |

# [8] Holes - Fiducials

DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V
(fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

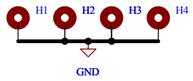
DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V

**LAYOUT NOTE:**
**blabla**

Description

Vout≈Vref+(Gain)×(I×Rshunt)

Shouse your INA240A...



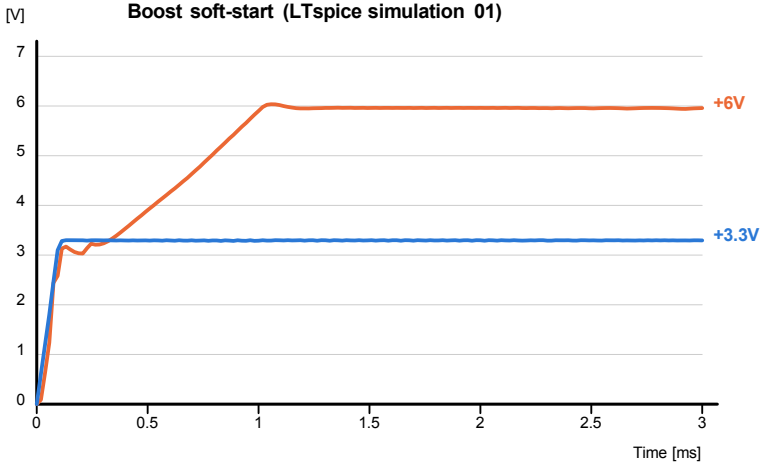

INA240 + Shunt

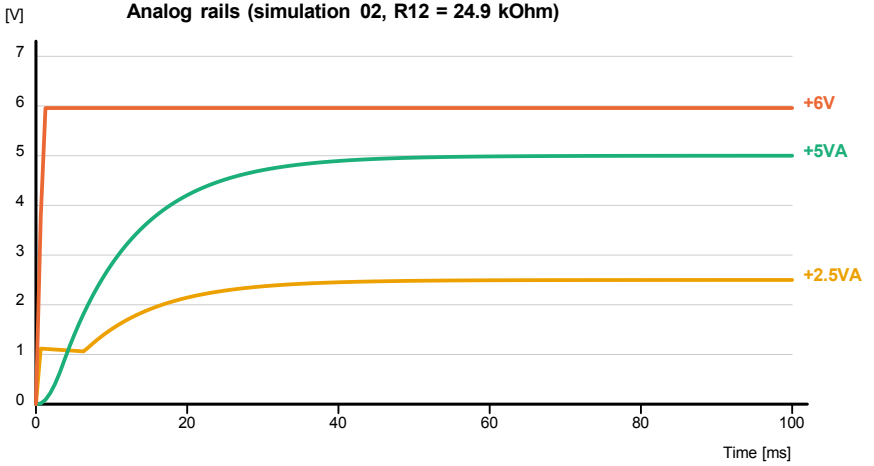
FILTRE

Test Point

| | | | | |
|---|---|---|---|---|
| Comments: **References:** Flexible I/O worked examples; Flexible I/O source configuration | Company: **OPEN THRUST LAB** | | Variant: | Git Hash: 7703dc5 |
| | Board Name: **PHASETRACKER** | | Project Name: **OPEN_PHASE** | |
| Sheet Title: Holes - Fiducials | File Name: Holes - Fiducials.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-05 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Holes, Fiducials/ | | Reviewer: | Size: **A3** | Sheet: **8** of **10** |

# [9] Power - Sequencing




| NAME | SOURCE | LEVEL | TIMING (after +3.3V is applied) |
|---|---|---|---|
| +3.3V | Host (no connector on the board yet) | 3.3 V | t = 0 |
| +6V | IC5 LT8362 boost, 2 MHz, soft-start C24 = 10 nF | 5.96 V | 90 % after 0.9 ms |
| +5VA | IC7 LTC6655-5 -> R8 -> SET of IC8 LT3045-1 | 5.000 V | 1 % after 48 ms, 0.1 % after 71 ms (tau = 9.8 ms) |
| +2.5VA | IC3 LTC6655-2.5 -> R11 -> SET of IC9 LT3045-1 | 2.500 V (*) | same time constant as +5VA |

(*) As built R12 = 49.9 kOhm gives 2.549 V. The curve and the level are given for R12 = 24.9 kOhm. See README, DESIGN REVIEW 1.

No Power-Good signal (PGFB tied to IN): wait at least 100 ms after power-up before the first conversion.

| | | | | |
|---|---|---|---|---|
| Comments: | Company: OPEN THRUST LAB | | Variant: | Git Hash: 7703dc5 |
| | Board Name: PHASETRACKER | | Project Name: OPEN_PHASE | |
| Sheet Title: Power - Sequencing | File Name: Power - Sequencing.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-06 | Revision: + (Unreleased) |
| Sheet Path: /Power - Sequencing/ | | Reviewer: | Size: A4 | Sheet: 9 of 10 |

# [10] Revision History

**Version Unreleased**

---

Added

- 3x PBV 1 mOhm shunt + INA241A2 (20 V/V)
- 3x phase voltage divider 40.2k / 10k
- AD7606C-18 ADC, 6 inputs used
- +3.3V -> LT8362 6 V -> LT3045-1 +5VA / +2.5VA
- Self-contained libs and 3D models (lib/)
- LTspice simulations (Simulation/LTspice)
- Predicted measurement quality (sim 07, 08)

---

Changed

- Renamed to OPEN_PHASE, KiCad 10
- Sheets named after their function
- IC3: LTC6655LN-2.5 -> LTC6655-2.5 (pin 6 = VOUT_S)
- R12: 49.9k -> 24.9k (+2.5VA = 2.500 V)

| | | | | | |
|---|---|---|---|---|---|
| | Comments: | Company: OPEN THRUST LAB | | Variant: | Git Hash: 7703dc5 |
| | | Board Name: PHASETRACKER | | Project Name: OPEN_PHASE | |
| | Sheet Title: Revision History | File Name: Revision History.kicad_sch | Designer: SIEBERT DIMITRY | Date: 2026-10-05 | Revision: + (Unreleased) |
| | Sheet Path: /Revision History/ | | Reviewer: | Size: A4 | Sheet: 10 of 10 |